

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

PDS Practical - V

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Class :SYIT	Batch : 1
Date of Assignment :12/08/26	Date/Time of Submission :12/08/26

Part – A Theory

5a) Write a python code to demonstrate importing pandas libraries and create frame data object.

Code:

```

import pandas as pd

data = {
    "name": ["Sally", "Mary", "John"],
    "age": [50, 40, 30],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)

newdf = df.filter(items=["name", "age"])
print(df)

```

Output:

```

   name  age  qualified
0  Sally   50      True
1  Mary   40     False
2  John   30     False

```

5b) Write a python code to show stastical information on given data set.
Code:

```
import statistics
# Calculate harmonic mean
print(statistics.harmonic_mean([40, 60, 80]))
print(statistics.harmonic_mean([10, 30, 50, 70, 90]))
print("mean")
# Calculate middle values
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print("median")
# Calculate the mode
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
print("mode")
# Calculate the standard deviation from a sample of data
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print("std dev")
# Calculate the variance from a sample of data
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print("variance")
print("Khan Shagufta S018")
```

Output:

```
55.38461538461538
27.97513321492007
mean
7
6.0
median
1
red
mode
3.7416573867739413
0.6925797186365383
std dev
14
0.4796666666666663
variance
Khan Shagufta S018
```

5c) Write a python code to create pandas series from dictionaries.<https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/>
Code:

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'A': 10, 'B': 20, 'C': 30}

# create a series
series = pd.Series(dictionary)

print(series)
print("Khan Shagufta S018")
|
```

Output:

```
A    10
B    20
C    30
dtype: int64
Khan Shagufta S018
|
```

5d)Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

```
import pandas as pd

data = {
    "Name": ["Sidra", "Aisha", "Iqra", "Firdos", "Alisha"],
    "Age": [22, 20, 17, 15, 18],
    "Hobby": ["dance", "singing", "cricket", "Drawing", "swimming"]
}

df = pd.DataFrame(data)
newdf = df.filter(items=["Name", "Age"])

print(newdf)
```

Output:

	Name	Age
0	Sidra	22
1	Aisha	20
2	Iqra	17
3	Firdos	15
4	Alisha	18